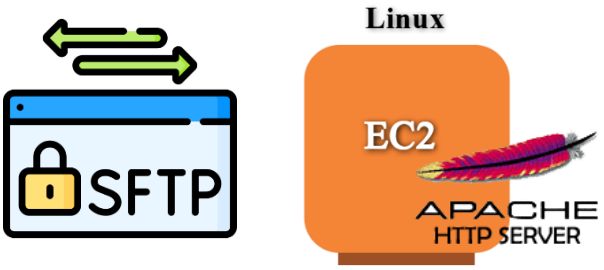
AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

Trainer: Aravindraj.G- Nminds Academy

AWS Academy Certified Educator

7.Configure SFTP to Linux Web Server in AWS



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Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

Use Cases for EC2 Linux Instances:

- Web and Application Hosting: Run Apache web servers or other Linux-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Linux instances, using SSH or EC2 for more control.
- Remote Desktop Applications: Set up SSH (Remote Desktop Services) for remote access to Linux desktops in the cloud.
- Development and Testing: Create development environments with linux OS to test php,MySql or other Linux-based applications.

SFTP- Secure File Transfer Protocol

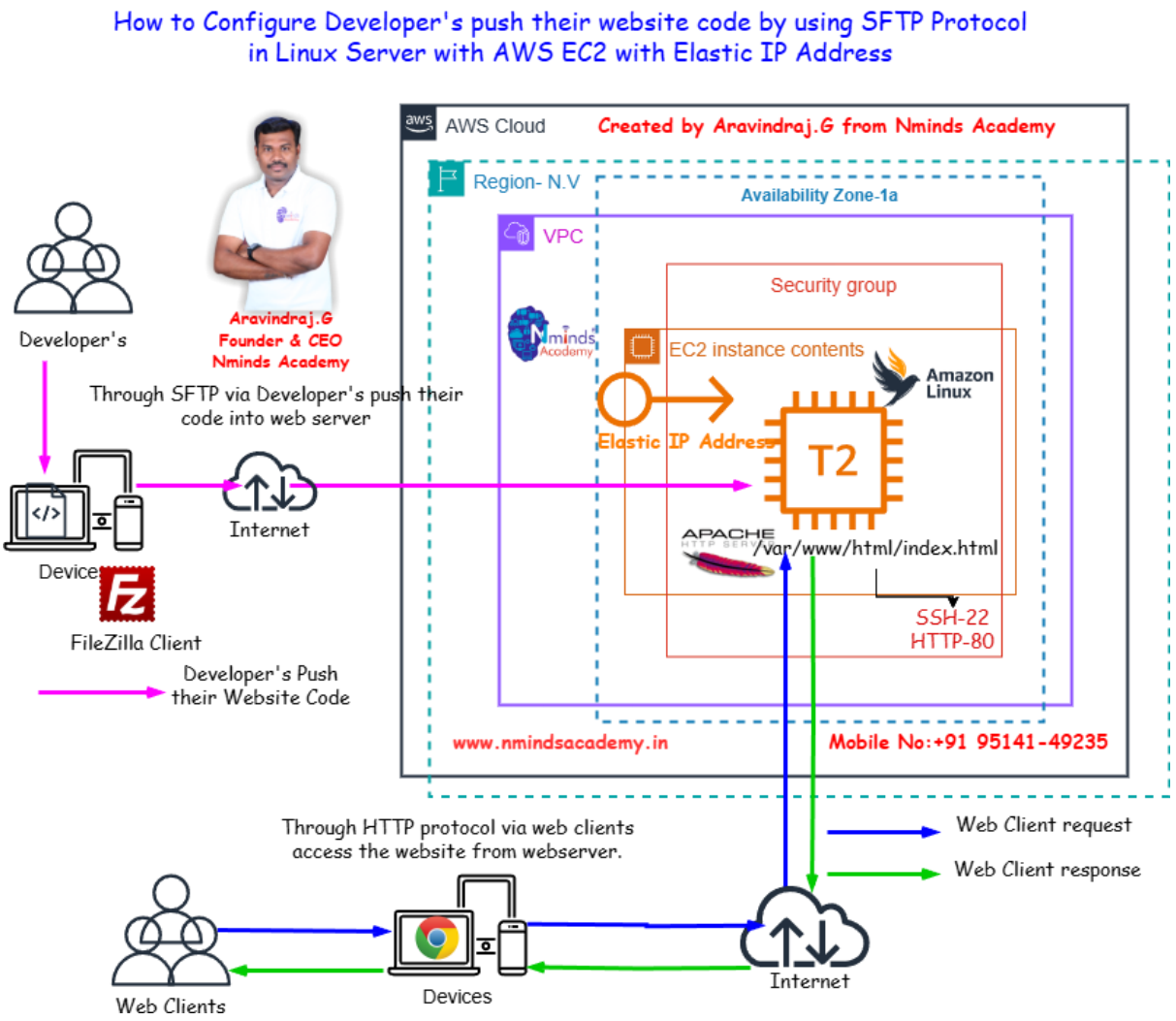
SFTP (Secure File Transfer Protocol) is a secure network protocol that allows users to transfer files over a secure, encrypted connection. It is a more secure alternative to FTP (File Transfer Protocol) and is commonly used to upload, download, and manage files on remote servers.

Here are the key uses and advantages of SFTP:

1. File Transfer: SFTP allows secure transfer of files between a local and a remote computer over a network. It ensures that the data is encrypted during transmission, protecting it from eavesdropping or tampering.
2. File Management: Besides transferring files, SFTP also allows users to perform file management tasks such as renaming, deleting, and moving files on the remote server.
3. Secure Authentication: SFTP uses SSH (Secure Shell) for authentication, which provides a stronger level of security compared to FTP's traditional username/password system. It often supports public key authentication, adding an extra layer of security.
4. Encryption: The main advantage of SFTP over FTP is encryption. All data sent over SFTP is encrypted, ensuring that sensitive information (like passwords and files) is protected during transmission.

5. Firewall-Friendly: SFTP operates over a single port (usually port 22), making it easier to use behind firewalls and NAT (Network Address Translation) devices compared to FTP, which uses multiple ports.

Topology



Used Services in AWS

Used Services in Windows Server

Amazon EC2

APACHE HTTP SERVER

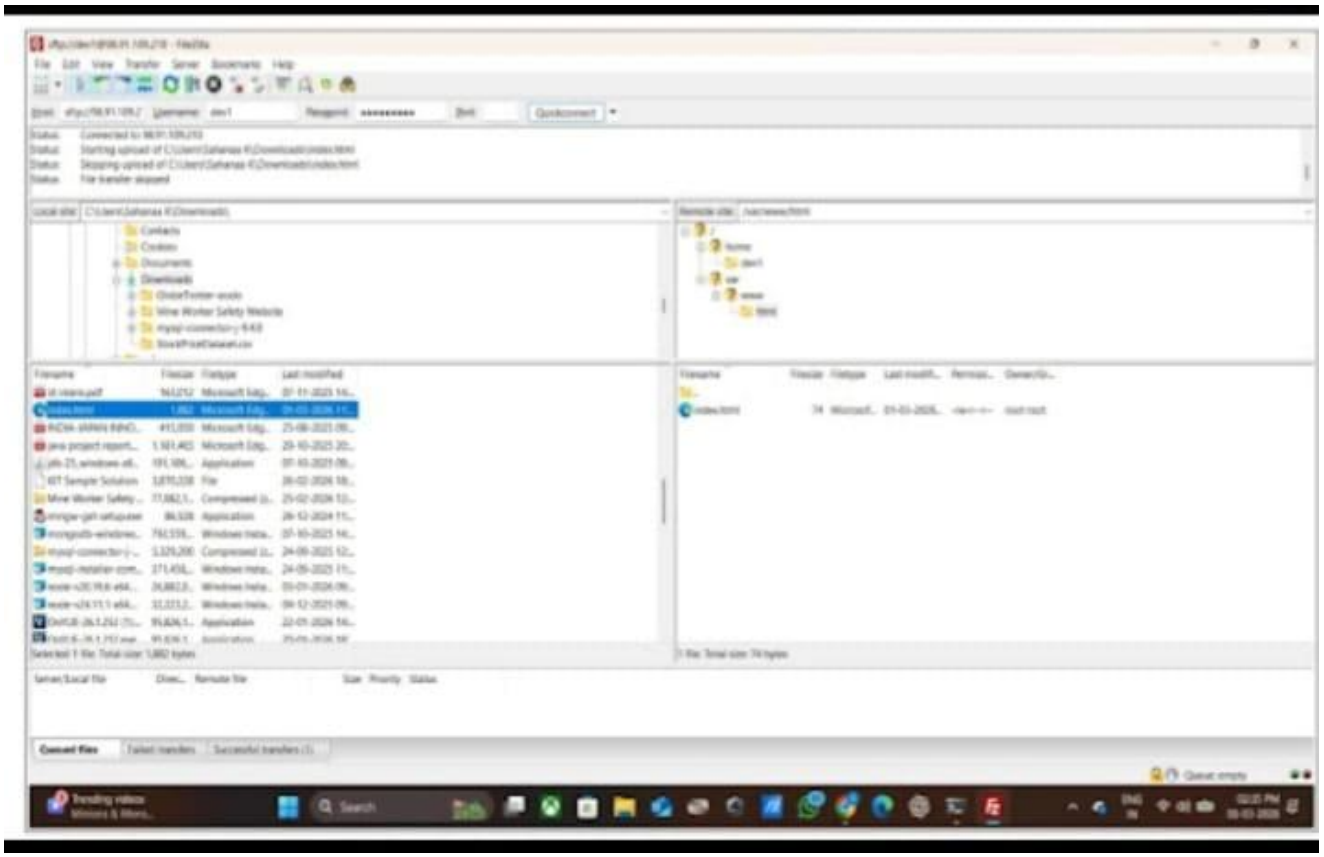
Apache Webserver

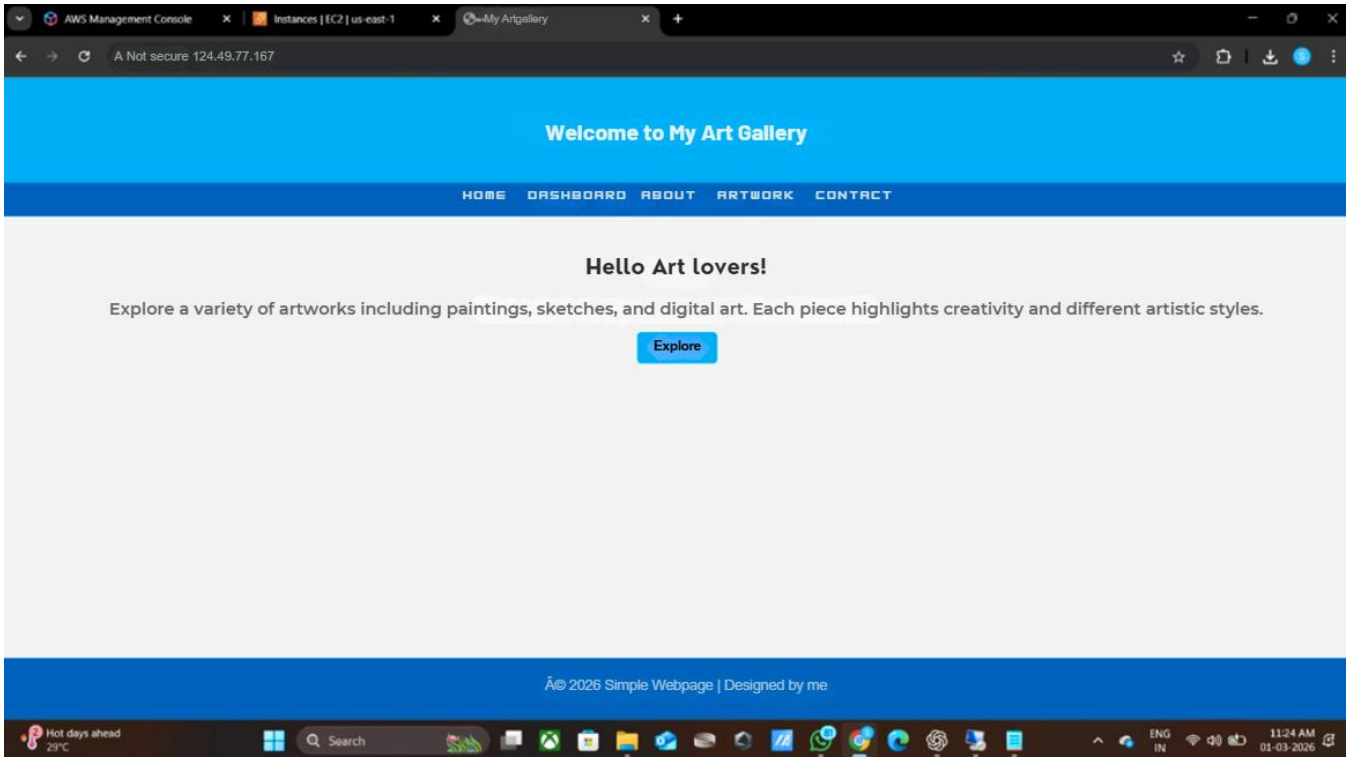
Execution Tasks:

```
================================================================================================================
 Package                        Architecture       Version                              Repository              Size
================================================================================================================
Installing:
 httpd                          x86_64             2.4.66-1.amzn2023.0.1                amazonlinux             47 k
Installing dependencies:
 apr                            x86_64             1.7.5-1.amzn2023.0.4                 amazonlinux            129 k
 apr-util                       x86_64             1.6.3-1.amzn2023.0.2                 amazonlinux             97 k
 apr-util-lmdb                  x86_64             1.6.3-1.amzn2023.0.2                 amazonlinux             13 k
 generic-logos-httpd            noarch             18.0.0-12.amzn2023.0.3               amazonlinux             19 k
 httpd-core                     x86_64             2.4.66-1.amzn2023.0.1                amazonlinux            1.4 M
 httpd-filesystem               noarch             2.4.66-1.amzn2023.0.1                amazonlinux             13 k
 httpd-tools                    x86_64             2.4.66-1.amzn2023.0.1                amazonlinux             81 k
 libbrotli                      x86_64             1.0.9-4.amzn2023.0.2                 amazonlinux            315 k
 mailcap                        noarch             2.1.49-3.amzn2023.0.3                amazonlinux             33 k
Installing weak dependencies:
 apr-util-openssl               x86_64             1.6.3-1.amzn2023.0.2                 amazonlinux             15 k
 mod_http2                      x86_64             2.0.27-1.amzn2023.0.3                amazonlinux            166 k
 mod_lua                        x86_64             2.4.66-1.amzn2023.0.1                amazonlinux             60 k

Transaction Summary
================================================================================================================
Install  13 Packages

Total download size: 2.4 M
Installed size: 6.9 M
Downloading Packages:
(1/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm                                  2.6 MB/s |  97 kB     00:00
(2/13): apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64.rpm                             332 kB/s |  13 kB     00:00
(3/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm                                       2.7 MB/s | 129 kB     00:00
(4/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm                          551 kB/s |  15 kB     00:00
(5/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm                     642 kB/s |  19 kB     00:00
(6/13): httpd-2.4.66-1.amzn2023.0.1.x86_64.rpm                                    1.5 MB/s |  47 kB     00:00
(7/13): httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch.rpm                         516 kB/s |  13 kB     00:00
(8/13): httpd-tools-2.4.66-1.amzn2023.0.1.x86_64.rpm                              2.9 MB/s |  81 kB     00:00
(9/13): httpd-core-2.4.66-1.amzn2023.0.1.x86_64.rpm                                27 MB/s | 1.4 MB     00:00
(10/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm                                9.6 MB/s | 315 kB     00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm                                 1.1 MB/s |  33 kB     00:00
(12/13): mod_http2-2.0.27-1.amzn2023.0.3.x86_64.rpm                               6.0 MB/s | 166 kB     00:00
(13/13): mod_lua-2.4.66-1.amzn2023.0.1.x86_64.rpm                                 1.9 MB/s |  60 kB     00:00
----------------------------------------------------------------------------------------------------------------
```

```
Total                                                                                                    12 MB/s | 2.4 MB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                                                          1/1
  Installing       : apr-1.7.5-1.amzn2023.0.4.x86_64                                                                         1/13
  Installing       : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64                                                               2/13
  Installing       : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64                                                            3/13
  Installing       : apr-util-1.6.3-1.amzn2023.0.2.x86_64                                                                    4/13
  Installing       : mailcap-2.1.49-3.amzn2023.0.3.noarch                                                                    5/13
  Installing       : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                                                                6/13
  Installing       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                                                   7/13
  Running scriptlet: httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch                                                           8/13
  Installing       : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch                                                           8/13
  Installing       : httpd-core-2.4.66-1.amzn2023.0.1.x86_64                                                                 9/13
  Installing       : mod_http2-2.0.27-1.amzn2023.0.3.x86_64                                                                 10/13
  Installing       : mod_lua-2.4.66-1.amzn2023.0.1.x86_64                                                                   11/13
  Installing       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                                                      12/13
  Installing       : httpd-2.4.66-1.amzn2023.0.1.x86_64                                                                     13/13
  Running scriptlet: httpd-2.4.66-1.amzn2023.0.1.x86_64                                                                     13/13
  Verifying        : apr-1.7.5-1.amzn2023.0.4.x86_64                                                                         1/13
  Verifying        : apr-util-1.6.3-1.amzn2023.0.2.x86_64                                                                    2/13
  Verifying        : apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64                                                               3/13
  Verifying        : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64                                                            4/13
  Verifying        : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch                                                       5/13
  Verifying        : httpd-2.4.66-1.amzn2023.0.1.x86_64                                                                      6/13
  Verifying        : httpd-core-2.4.66-1.amzn2023.0.1.x86_64                                                                 7/13
  Verifying        : httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch                                                           8/13
  Verifying        : httpd-tools-2.4.66-1.amzn2023.0.1.x86_64                                                                9/13
  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64                                                                  10/13
  Verifying        : mailcap-2.1.49-3.amzn2023.0.3.noarch                                                                   11/13
  Verifying        : mod_http2-2.0.27-1.amzn2023.0.3.x86_64                                                                 12/13
  Verifying        : mod_lua-2.4.66-1.amzn2023.0.1.x86_64                                                                   13/13

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64                          apr-util-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64                apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch        httpd-2.4.66-1.amzn2023.0.1.x86_64
  httpd-core-2.4.66-1.amzn2023.0.1.x86_64                  httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch
```

```
Complete!
[root@ip-172-31-8-247 ~]# systenmctl status httpd
-bash: systenmctl: command not found
[root@ip-172-31-8-247 ~]# systemctl status httpd
○ httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: inactive (dead)
       Docs: man:httpd.service(8)
[root@ip-172-31-8-247 ~]# systemctl start httpd
[root@ip-172-31-8-247 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Sun 2026-03-01 06:04:41 UTC; 12s ago
       Docs: man:httpd.service(8)
   Main PID: 25513 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1067)
     Memory: 13.3M
        CPU: 68ms
     CGroup: /system.slice/httpd.service
             ├─25513 /usr/sbin/httpd -DFOREGROUND
             ├─25514 /usr/sbin/httpd -DFOREGROUND
             ├─25515 /usr/sbin/httpd -DFOREGROUND
             ├─25516 /usr/sbin/httpd -DFOREGROUND
             └─25517 /usr/sbin/httpd -DFOREGROUND

Mar 01 06:04:41 ip-172-31-8-247.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Mar 01 06:04:41 ip-172-31-8-247.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Mar 01 06:04:41 ip-172-31-8-247.ec2.internal httpd[25513]: Server configured, listening on: port 80
[root@ip-172-31-8-247 ~]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-172-31-8-247 ~]#
```




AWS Management Console
Instances | EC2 | us-east-1
My Artgallery
A Not secure 124.49.77.167
Welcome to My Art Gallery
HOME DASHBOARD ABOUT ARTWORK CONTACT
Hello Art lovers!
Explore a variety of artworks including paintings, sketches, and digital art. Each piece highlights creativity and different artistic styles.
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